4

States of matter

THE most common reactions happen in solution and involve cations and anions. Think, for example, when you add sugar to your coffee or how metal rusts when it gets wet. The first example involves a dissolution reaction, whereas the second a redox reaction in which electrons are being exchanged. Overall there are three main different types of reactions happening in solutions. It is not only important to be able to differentiate these reactions, but it is also important to understand the nuances between acid-base reactions, precipitation reactions, and redox reactions. It is also critical to understand the properties of solutions and learn how to quantify and compare the amount of solute in a solution. On one hand, most of you will be surprised to know that water does not conduct electricity. This is because pure water is a weak electrolyte. On the other hand, the importance of electrolytes is well known among the sports community. If you have ever played a sport, you have probably chugged a sports drink. These are electrolyte solutions. However, few know the specifics of their function. Electrolytes are salts that conduct electricity in water by separating into positive and negative ions. To understand the properties of solutions it is important to identify and differentiate the different types of electrolytes.

4.1 Ions & ionic charges

`Mass percent concentration` The mass percent (m/m) is the amount of solute in grams per grams of solution in percent form

$$\boxed{m/m = \frac{\text{g of solute}}{\text{g of solution}} \times 100}$$

Sample Problem 1

A NaCl solution is prepared by mixing 4g of NaCl with 50g of H_2O. Calculate the percent (m/m) of the solution.

SOLUTION

We need the grams of solute and the grams of solution. The grams of solute are given (4g of NaCl), whereas the grams of solution result from adding the grams of solvent and solute: 54 g of solution. Using the formula for the percent (m/m), he have:

$$m/m = \frac{\text{4 g of solute}}{\text{54 g of solution}} \times 100 = 7.4\%$$

This means that by mixing 4g of NaCl with 50g of H_2O you prepare a 7.4% (m/m) solution.

STUDY CHECK

A KCl solution is prepared by mixing 5g of KCl with 200g of H_2O. Calculate the percent (m/m) of the solution.

Volume percent concentration The volume percent concentration (v/v) is the volume of solute per volume of solution in percent form

$$\boxed{v/v = \frac{\text{volume of solute}}{\text{volume of solution}} \times 100}$$

Mass/volume percent concentration The mass/volume percent concentration (m/v) is the mass of solute per mL of solution in percent form.

$$\boxed{m/v = \frac{\text{g of solute}}{\text{mL of solution}} \times 100}$$

Molarity concentration The molarity (M) is the moles of solute per L of solution.

$$\boxed{M = \frac{\text{moles of solute}}{\text{L of solution}}}$$

Sample Problem 2

A NaCl solution is prepared by mixing 4g of NaCl (MW=58.4g/mol) with 50 g of water until a final volume of 52mL of solution. Calculate: (a) the mass percent (m/m) concentration; (b) the molarity.

SOLUTION

(a) to calculate the mass percent (m/m) we just need the grams of solute and the grams of solution–that is four plus fifty. Both numbers are already given:

$$m/m = \frac{\text{g of solute}}{\text{g of solution}} \times 100 = \frac{\text{4 g of solute}}{\text{54 g of solution}} \times 100 = 9.2\%$$

(b) To calculate molarity we need the moles of solute and the liters of solution. We have the mL of solution, that can be converted to L: 52mL= $5.2 \times 10^{-2} L$. To calculate the moles of solute, we will use the grams of solute and the molar mass to convert this value into moles: $4g/58.4g/mol$ =0.068moles. Plugging all values into the molarity formula:

$$M = \frac{\text{moles of solute}}{\text{L of solution}} = \frac{\text{0.068 moles of solute}}{5.2 \times 10^{-2}\text{L of solution}} = 1.31M$$

STUDY CHECK

(a) A KCl solution is prepared by mixing 8g of NaCl (MW=74g/mol) with 250mL of H_2O. Calculate the molarity; (b) A KCl solution is prepared by mixing 5g of KCl with 200g of H_2O. Calculate the percent (m/m) of the solution.

Concentration units as conversion factors Each of the different concentration units–molarity, mass percent, volume percent, mass/volume percent–can be

used in a conversion factor form. For example, if the molarity of a solution is 3M, this means that in the solution there is 3 moles of solute every one litter of solution.

$$\boxed{3\text{M} \quad \text{or} \quad \frac{3 \text{ mol of solute}}{1 \text{ L of solution}} \quad \text{or} \quad \frac{1 \text{ L of solution}}{3 \text{ mol of solute}}}$$

Similarly, if the mass percent of a solution is 5% this means that there is 5 grams of solute every 100 grams of solution. We often use concentration units as conversion factors when we need to transform between on unit on top (bottom) of the conversion factor and the unit on the bottom (top).

Sample Problem 3

How much volume of a 4M solution do you need to provide 5 moles of solute.

SOLUTION

We will use the conversion factor of Molarity using the volume on top and the moles on the bottom in order to cancel the units:

$$5\cancel{\text{moles of solute}} \times \frac{1 \text{ L of solution}}{4\cancel{\text{moles of solute}}} = 1.25\text{L}$$

This means that 1.25L of a 4M solution will provide 5 moles of solute.

STUDY CHECK

How many grams of a 6% (m/m) solution do you need to provide 5 grams of solute.

`Dilution` Dilution is the process for preparing a diluted solution from a more concentrated solution. Solutions are often times stored in a stock room in concentrated form. These stocks should be diluted before use. In order to dilute a solution we need to take a certain amount of the concentrated solution and add water. When adding water, the number of moles of solute does not change, and the concentration always decreases. We have a concentrated solution (c_1) and we need to prepare a certain volume (V_2) of a more diluted solution (c_2). The question is how much volume of the concentrated solution (V_1) we need to take. In order to answer this we should use the following formula:

$$\boxed{c_1 \cdot V_1 = c_2 \cdot V_2}$$

Sample Problem 4

How many liters of a 3M NaCl solution are required to prepare 2L of a 1M diluted NaCl solution.

SOLUTION

We have a concentrated solution of 3M molarity and we want to prepare a more dilute solution. In particular 2L of a 1M. Hence: $c_1 = 3$ and $c_2 = 1M$ and $V_2 = 2L$. Using the dilution formula:

$$3M \cdot V_1 = 1M \cdot 2L$$

Solving for V_1 we have a volume of 0.66L.

STUDY CHECK

How many liters of a 5M NaCl solution are required to prepare 3L of a 3M diluted NaCl solution.

4.2 The mole

Some of the terms you use in your everyday life actually refer to a number. For example, you buy a pair of socks–two socks–or you buy a dozen of eggs from the grocery store–twelve eggs–and sometimes you buy a case of beers–24 cans. In a chemistry laboratory we normally do not weight small numbers of molecules of a chemical. In chemistry, molecules are counted by the mole, and the term mole–abbreviated as mol–refers to the6.022×10^{23} number. For example, a mol of CO molecules contains 6.022×10^{23} molecules of CO, and a mol of water molecules contains 6.022×10^{23} molecules of water. This is because the word mole means the number 6.022×10^{23}, similarly as the word pair means the number two. The number 6.022×10^{23} is called Avogadro's number, in reference to Amedeo Avogadro, the Italian physicists who coined the term. In chemistry labs, chemicals are often measured by weight. In this section we will show how to convert moles into weight–into grams–by using a property called the molecular weight. Finally, mind that the term mol and molecule even if they look similar they are not. Molecule refers to a combination of atoms and mole refers to a large number of molecules. As a note, the abbreviation of mole is mol, and for example we will say seven mol of H_2O.

From moles to molecules One mole of molecules contains 6.022×10^{23} molecules. This is because the term mole refers to Avogadro's number. Hence we can use the follow unit equivalency:

$$\boxed{1\text{mol of } H_2O = 6.02 \times 10^{23}\text{molecules of } H_2O}$$

or a conversion factor to transform moles into molecules or molecules into moles as well:

$$\boxed{\frac{1\text{mol of } H_2O}{6.02 \times 10^{23}\text{molecules of } H_2O} \text{ or } \frac{6.02 \times 10^{23}\text{molecules of } H_2O}{1\text{mol of } H_2O}}$$

For example: how many molecules are there in 3 mol of H_2O? In order to calculate the number of molecules of H_2O in 3 moles of H_2O, (moles $\rightarrow$ molecules) you need to set up a conversion factor, starting by the give information (3 moles) and using the mol-to-molecule conversion factor with mol on the bottom:

$$3\cancel{\text{moles of } H_2O} \times \frac{6.02 \times 10^{23} \text{ molecules of } H_2O}{1\cancel{\text{mol of } H_2O}} = 1.80 \times 10^{24}\text{molecules of } H_2O$$

In you need to convert molecules to moles (molecules $\rightarrow$ moles), you just need to follow the same procedure, using the conversion factor between mol-to-molecule with molecule in the bottom. For example 3×10^{20} H_2O molecules equals to 4.98×10^{-4}moles of H_2O as

$$3 \times 10^{20}\cancel{\text{molecules of } H_2O} \times \frac{1\text{moles of } H_2O}{6.02 \times 10^{23}\cancel{\text{molecules of } H_2O}} = 4.98 \times 10^{-4}\text{moles of } H_2O$$

From molecules to atoms Molecules are made of atoms, and for example the CO_2 molecule contains an atom of C and two atoms of O. In order to convert from molecules to atoms (molecules $\rightarrow$ atoms) you need to use the coefficients in the molecular formula. For example, and H_2O molecule contains an atom of O and two atoms of H, and hence the relation between water molecules and H and O atoms is:

$$\boxed{\frac{1 \text{ molecule of } H_2O}{1 \text{ atom of O}} \text{ and } \frac{1 \text{ molecule of } H_2O}{2 \text{ atom of H}}}$$

To convert from moles into atoms (moles $\rightarrow$ atoms) you need to use a two-step process in a single line. First you convert from moles into molecules, to then convert molecules into atoms. For example, 3 moles of H_2O contains 1.6×10^{24} H atoms, as:

$$3\,\cancel{\text{moles of } H_2O} \times \frac{6.02 \times 10^{23}\ \cancel{\text{molecules of } H_2O}}{1\,\cancel{\text{moles of } H_2O}} \times \frac{2\text{H atoms}}{1\ \cancel{\text{molecules of } H_2O}}$$
$$= 1.6 \times 10^{24}\text{atoms of H}$$

Sample Problem 5

Calculate: (a) the number of CuO molecules in 3.4 moles of CuO; (b) the number of moles of CO in 5×10^{20} CO molecules

SOLUTION

(a) 3.4 moles of CuO equals to 2.05×10^{22} molecules of CuO as:

$$3.4\,\cancel{\text{moles of CuO}} \times \frac{6.022 \times 10^{23}\text{ molecules of CuO}}{1\,\cancel{\text{mole of CuO}}} =$$
$$= 2.05 \times 10^{22}\text{molecules of CuO}$$

(b) 5×10^{20} CO molecules equals to 8.3×10^{-4} moles of CO, as

$$5 \times 10^{20}\,\cancel{\text{CO molecules}} \times \frac{1\text{ mole of CO}}{6.022 \times 10^{23}\ \cancel{\text{CO molecules}}} = 8.3 \times 10^{-4}\text{moles of CO}$$

STUDY CHECK

The chemical formula for caffeine is $C_8H_{10}N_4O_2$. Calculate the number of C, H, N and O atoms in 3.5 moles of caffeine.

4.3 Converting moles into grams and into atoms

A standard way to measure chemicals in the lab is by weight. We can weight different quantities and the larger the quantity the larger the weight. For a chemical, the weigh of a mole is called the molar (or molecular) weight. For example if we weight a mole of water (H_2O) we will be weighting 18 grams of water, or if you weight a mole of table salt (NaCl) the scale will show 58 grams. In this section you will learn how to calculate the molar mass of a chemical and how to use this property to convert from weight to moles (and moles to weight).

Molar mass of a chemical Chemicals are made of atoms, and each atom has an specific atomic weight (AW) listed in the periodic table. For example, the atomic weight of Na is 23 grams whereas the atomic weight of Cl is 35 g. The weight of all the atoms of a molecule is called the molecular weight (we call this also molar weight or MW). For example, the molecular weight of NaCl is 58 g, as the weight of Na and Cl is 23 and 35g. Another example would be water, H_2O with a molecular weight of 18g–as the atomic weight of H and O is 1 and 16 g, respectively, and the molecule has two H atoms. The units for molecular weight is $\frac{g}{mol}$, also written as g/mol. In order to compute the molar mass of a molecule you need to break down the molecule into atoms using the coefficients in the formula. For example, the formula for vinegar is $C_2H_4O_2$ that means a vinegar molecule contains 2C, 4H and 2O atoms. If you add the atomic masses of 2C, 4H and 2O you will get $60g/mol$. If the chemical formula has a parenthesis, you

need to open up the parenthesis to calculate the total number of atoms. As an example, $Ca(NO_3)_2$ contains 1Ca, 2N, and 6O, and its molar mass is $164.09 g/mol$.

Sample Problem 6

Calculate: (a) The atomic weight of Mg; (b) the molecular mass of sulfuric acid, H_2SO_4

SOLUTION

(a) According to the periodic table the atomic weight (AW) of Mg is $24.31 g/mol$. (b) The molar mass of H_2SO_4 is the result of adding the atomic masses of 2H (AW=$1 g/mol$) atoms, 1 S (AW=$32 g/mol$) and 4O (AW=$16 g/mol$) atoms, that gives $98.08 g/mol$.

STUDY CHECK

Calculate the molar mass of glucose $C_6H_{12}O_6$

`From moles to grams` The molar mass is used to convert moles to grams or grams to mol. For example, the molar mass of water is $18 g/mol$. This means:

$$\boxed{1\text{mole of } H_2O = 18\text{g of } H_2O}$$

that is the same as

$$\boxed{\frac{1 \text{ mole of } H_2O}{18 \text{ grams of } H_2O} \text{ or } \frac{18 \text{ grams of } H_2O}{1 \text{ mole of } H_2O}}$$

Sample Problem 7

Smelling salts ($(NH_4)_2CO_3$) are chemical used to arouse consciousness. These are used by pro athletes to get into the zone before a game. How many moles of salt do you have in 100 grams of these salts?

SOLUTION

We first need to calculate the molar mass of $(NH_4)_2CO_3$, a chemical with 2N, 8H, 1C and 3O atoms. The molar mass hence would be: $2 \times 5 + 8 \times 1 + 1 \times 12 + 3 \times 16 = 96 g/mol$. In order to calculate the moles given the gram, you need to use the molar mass a conversion factor:

$$100 \cancel{\text{g of } (NH_4)_2CO_3} \times \frac{\text{moles of } (NH_4)_2CO_3}{96 \cancel{\text{ g of } (NH_4)_2CO_3}} = 1.04 \text{ moles of } (NH_4)_2CO_3$$

STUDY CHECK

Calculate the MW of table salt (NaCl) and the grams in 20 moles of this salt.

4.4 Radiation measurement, units and radiation effects

Beta and gamma radiation can be detected with a Geiger counter, which consists of a detector tube with a specific ionizing gas. When radiation enters the Geiger counter, it generates charged particles that produce a detectable electrical current. The larger the current the stronger the radioactive source. In the following we will address the different

units of radioactivity–activity, adsorbed dose and biological damage–as well as the effects of radiation.

`Activity units` The radioactivity of an isotope, often referred as *activity*, can be measure in two different units: Curies (Ci) or becquerel (Bq). Curie was somehow the original unit employed to measure the radioactivity of radium and becquerel is a more modern unit of radioactivity. Bq is the SI unit of activity. Both units are related by:

$$\boxed{1Ci = 3.7 \times 10^{10} Bq} \quad \text{or} \quad \boxed{\frac{1Ci}{3.7 \times 10^{10} Bq}} \quad \text{or} \quad \boxed{\frac{3.7 \times 10^{10} Bq}{1Ci}} \tag{4.1}$$

Activity refers to the isotope and is also measured in disintegration per minute (cpm).

`Adsorbed dose` Whereas activity refers to the isotope, the adsorbed dose refers to the body that receives radiations. The unit for adsorbed dose is call rad (radiation adsorbed dose).This unit refers to the amount of radiation adsorbed per gram of material. The SI unit for adsorbed dose is the gray (Gy).

`Radiation equivalent in humans, rem` Not all radiations have the same impact on the human body. The radiation equivalent in humans takes into account the different types of radiation in order to adjust the biological damage of radiations. the rem is the number of rads times a factor that depends on the radiation. This factor is one for beta and gamma radiation, being 20 for alpha particles.

`Exposure to radiation` We are all somehow exposed to radiation every day. The reason for this background radiation is that there are many natural radioisotopes that form the atoms of many materials such as brick, concrete, water or even the air. Still, the daily exposure is very low and you should not be concern by the effect of this background radiation.

Table 4.1 Radiation Measurement

Activity	curie (Ci)
	Becquerel (Bq) = 2.7×10^{-11} Ci
	1Ci = 2.22×10^{12} dpm
Dosage (D)	rad
	gray (Gy) =100rad
Damage (H)	rem
	Sievert (Sv) = 100rem
rem=rad $\times$ Factor	

`Dangers of radiation` Radiation units different than Ci or Bq are used to measure the impact of radiation in humans. The rem (radiation equivalent in humans) is a radiation unit that measures the direct biological effects of different kinds of radiation. The amount of rems a person receives would determine the impact of the radiation on this person's health. As an example, radiation exposure under 25 rem are harmless and they cannot be detected. If a victim is exposed to 100 rem or higher, the person will

suffer the symptoms of radiation sickness, and will feel nausea, vomit, fatigue, and a reduction in white-cell count. If a person is exposed to a dosage greater than 300 rem, that can lower the white-cell count to zero; the victim will suffer diarrhea, hair loss, and infection. Exposure to radiation of about 500 rem is expected to cause death in half of the people receiving that dose. Radiation dosages of about 600 rem would be fatal to all humans within a few weeks.

Sample Problem 8

Ioflupane is a radiopharmaceutical that helps visualize the brain of Parkinson patients. A injection of this drug has a 5mCi activity. Convert this value to MBq.

SOLUTION

You need to use the conversion factor between Bq and Ci, $\frac{1Ci}{3.7\times10^{10}Bq}$, as well as the conversion factor between mCi and Ci $\frac{1mCi}{1\times10^{-3}Ci}$, and Bq and MBq $\frac{1MBq}{1\times10^{6}Bq}$.

$$5\cancel{mCi} \times \frac{1\times10^{-3}\cancel{Ci}}{1\cancel{mCi}} \times \frac{3.7\times10^{10}\cancel{Bq}}{1\cancel{Ci}} \times \frac{1MBq}{1\times10^{6}\cancel{Bq}} = 185MBq$$

STUDY CHECK

Quadramet is a radiopharmaceutical used treat pain when cancer has spread to the bone. A injection of this drug has a 740 MBq activity. Convert this value to mCi.

4.5 The PH scale

This section describes the PH scale that simply transforms a concentration value–often times a very small number–into a simple round value. In short, the PH value tells you how much protons are there in a solution so that the larger PH the fewer protons are there in solution. It also informs about the hydroxyl concentration, as protons and hydroxyls are connected by means of the dissociation equilibrium of water.

`Protons and Hydroxyls` Acids and bases exist on solution with water. That means that as they produce protons or hydroxyls water receives these ions as it ionizes as well. Hence, the concentration of protons and hydroxyls in solution are not independent. Indeed, the ion-product of water relates the concentration of protons ($[H^+]$) and the concentration of hydroxyls ($[OH^-]$):

$$[H^+]\cdot[OH^-] = 1.0\times10^{-14}$$

Water is neutral, which means that the concentration of protons ($[H^+]$) and the concentration of hydroxyls ($[OH^-]$) and both equal to $1.0\cdot10^{-7}$M. When we dissolve an acid or a base into water, $[OH^-]$ and $[H^+]$ change drastically. When dissolving an acid, $[H^+]$ increases as acids produce protons, while $[OH^-]$ decreases. Differently, when dissolving a base, $[OH^-]$ increases, as bases produce hydroxyls, while $[H^+]$ decreases.

Sample Problem 9

The proton concentration in an acid solution is $7.0 \cdot 10^{-5}$M. Calculate $[\mathrm{OH}^-]$.

SOLUTION

We will use Equation **??**. The value given is $[\mathrm{H}^+] = 7.0 \cdot 10^{-5}$M and the problem ask $[\mathrm{OH}^-]$. Solving for $[\mathrm{OH}^-]$ we have:

$$7.0 \cdot 10^{-5} \cdot [\mathrm{OH}^-] = 1.0 \cdot 10^{-14}$$

Hence $[\mathrm{OH}^-] = 1.4 \cdot 10^{-10}$M.

STUDY CHECK

The hydroxyl concentration in a basic solution is $2.3 \cdot 10^{-6}$M. Calculate the concentration of protons.

The PH scale The proton concentrations in aqueous solutions tend to be rather small. For example, the proton concentration in normal vinegar is $2 \cdot 10^{-3}$M. As it is hard to work with these small concentrations, scientists developed the PH scale that transforms $[\mathrm{H}^+]$ into a larger number. The formula for the PH is:

$$\boxed{PH = -log[\mathrm{H}^+]} \tag{4.2}$$

The PH scale normally ranges from 0 to 14. PH values lower than 7 correspond to acidic solutions, whereas PH values larger than 7 correspond to basic solutions. Solutions with PH of 7 are neutral. For example, the PH for vinegar is $-log(2 \cdot 10^{-3})$ that is 2.69. However, it exists PH values out of the scale for very concentrated solutions. Examples of PH values and common chemicals are given in the figure below. An equivalent scale is also defined for the concentration of hydroxyls. The POH values is defined as:

$$\boxed{POH = -log[\mathrm{OH}^-]} \tag{4.3}$$

The POH scale also ranges from 0 to 14. POH values lower than 7 correspond to this time to basic solutions, whereas POH values larger than 7 correspond to acidic solutions. Solutions with POH of 7 are neutral. The values of PH and POH are hence related by the following equation:

$$\boxed{PH + POH = 14} \tag{4.4}$$

For example, if the PH of a solution is 4 therefore the POH will be 10. Both indicators suggest that the solution would be acidic.

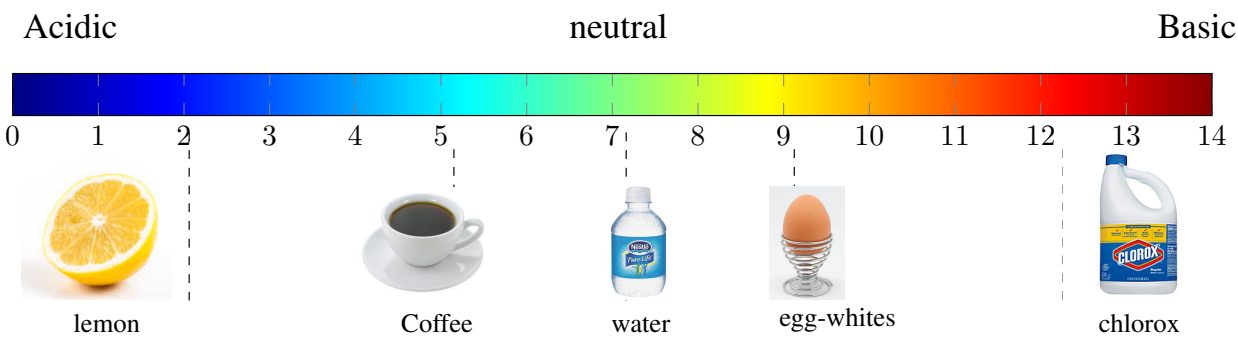


Figure 4.1 The PH scale

Sample Problem 10

Calculate the PH for: (a) an acid solution with proton concentration of $7.0 \cdot 10^{-5}$M (b) a basic solution with a hydroxyl concentration of $7.0 \cdot 10^{-5}$M.

SOLUTION

(a) We will use Equation 4.2. Given is $\left[\mathrm{H}^{+}\right] = 7.0 \cdot 10^{-5}$M and the problem ask for the PH. Solving for PH we have:

$$PH = -log\left(7.0 \cdot 10^{-5}\right)$$

and the results is 4.15. This is an acidic PH. (b) We will also use Equation 4.2. However, before doing that, we need to compute the concentration of protons. In order to do this we will use Equation **??** given $\left[\mathrm{OH}^{-}\right] = 8.0 \cdot 10^{-2}$M

$$\left[\mathrm{H}^{+}\right] \cdot 8.0 \cdot 10^{-2} = 1.0 \cdot 10^{-14}$$

We have $\left[\mathrm{H}^{+}\right] = 1.25 \cdot 10^{-13}$M. Now we can compute the PH. Solving for PH we have:

$$PH = -log\left(1.25 \cdot 10^{-13}\right)$$

and the results is 12.90. This is a basic PH.

STUDY CHECK

Calculate the PH for: (a) an acid solution with proton concentration of $3.0 \cdot 10^{-8}$M (b) a basic solution with a hydroxyl concentration of $2.0 \cdot 10^{-9}$M.

PK of an acid or base In the same way as we defined the PH and POH scale, we can also use the logarithmic notation to transform acidity and basicity constant into more manageable number. We define PK_a of an acid as:

$$\boxed{PK_a = -log\left(K_a\right)} \tag{4.5}$$

For example, as the acidity constant of acetic acid is 1.75×10^{-5} its PK_a would be 4.74. An equivalent definition exist for PK_b.

From PH to proton concentration At this point we know that the PH quantifies the proton concentration of a solution. So given $\left[\mathrm{H}^{+}\right]$ we can calculate PH by means of the logarithm with opposite sign. But what if we know the PH and we want to calculate the corresponding proton concentration? We can do this by using the formula:

$$\boxed{\left[\mathrm{H}^{+}\right] = 10^{-PH}} \tag{4.6}$$

In order to use the previous formula you need to use the power key in your calculator. For example if the PH is 3.3 and we need to calculate the proton concentration you will need to type: 10 $\boxed{\wedge}$ $\boxed{-}$ 3.3, and the result is $5.0 \cdot 10^{-4}$M. Mind that: (a) in some calculators, sometime the power key looks like: $\boxed{10^{x}}$; (b) you need to use the negative key and not the minus key. Minus is used for substations, the negative key is used for number. An equivalent relation exist between the concentration of hydroxyls and the POH:

$$\boxed{\left[\mathrm{OH}^{-}\right] = 10^{-POH}} \tag{4.7}$$

The diagram below displays some of the most important equations involved in this section:

Sample Problem 11

The PH of a solution is 4.5. Calculate the proton concentration of that solution.

SOLUTION

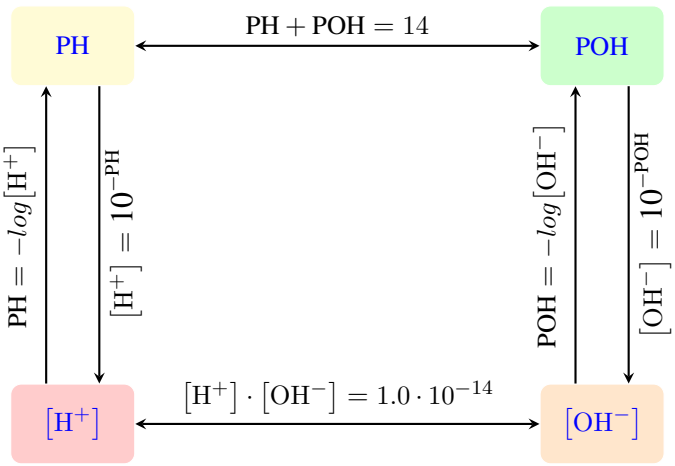
We will use Equation 4.6, given PH and asking $[H^+]$.

$$[H^+] = 10^{-PH} = 10^{-4.5}$$

and the results is $3.16 \cdot 10^{-5}$M.

STUDY CHECK

The PH of a solution is 9.5. Calculate the proton concentration of that solution.



4.6 Energy & temperature

When you are running, walking, dancing, or thinking, you are using energy to do work. In fact, energy is defined as the ability to do work. Suppose you are climbing a steep hill. Perhaps you become too tired to go on; you do not have sufficient energy to do any more work. Now suppose you sit down and have lunch. In a while you will have obtained energy from the food, and you will be able to do more work and complete the climb. Similarly, chemical energy is the energy stored in the structure of chemicals and it depends on the arrangement of molecules and the nature of these molecules.

`Potential & Kinetic Energy: heat` Energy can be classified as potential energy or kinetic energy. Kinetic energy is the energy of motion and any fast-moving object has kinetic energy. Think about a bullet coming out of a gun; as the bullet moves very fast it contains kinetic energy that can be releases when it collide with a target. Potential energy is energy stored in objects located at a certain height. A boulder resting on top of a mountain has potential energy because of its location. If the boulder rolls down the mountain, the potential energy becomes kinetic energy. Water stored in a reservoir has potential energy. When the water goes over the dam, the potential energy is converted to kinetic energy. The potential energy resulting from the interaction of charged particles is called electrostatic energy. Heat refers to thermal energy, which is associated with the random motion of particles in a substance and therefore is related to kinetic energy. A frozen pizza feels cold because the particles in the pizza are moving slowly. As the pizza receives heat, the motion of the particles increase. Eventually the

particles have enough energy to make the pizza hot and ready to be eaten. When a substance receives heat it gets warmer and it raises its temperature, whereas if it looses heat it gets cooler and its temperature decreases.

Mechanical energy: work The sum of potential and kinetic energy is called mechanical energy. Mechanical energy refers to the ability to do work. Examples of work are a car engine moving or a ballon expanding its volume.

The law of conservation energy In this chapter we will analyze energy changes associated with chemical reactions. In order to do this, we need to define the system and its surroundings. The system will be the chemical reaction often happening in a beaker, whereas the surroundings would be the area surrounding the beaker. The system plus its surroundings is called the universe. The beaker may lose energy, and in that case energy will flow from the system to the surroundings. Similarly, the system may gain energy, flowing from the surroundings to the beaker. In a closed system, the energy is being conserved and when one type of energy disappears, a different type of energy will appear. As an example, if you drop an object from the top of a building, originally the object had potential energy that converts into kinetic energy as the object gains speed. This is called the law of conservation of energy.

Energy units Two different units of energy are often employed: calories (cal) and joules (J). Joule is the SI unit of energy equal to $\mathrm{kg} \cdot \mathrm{m}^2/\mathrm{s}^2$. One can transform calories to joules and joules to calories using the following conversion factor:

$$\boxed{1\mathrm{cal} = 4.184\mathrm{J}} \quad \text{or} \quad \boxed{\frac{1\mathrm{cal}}{4.184\mathrm{J}}} \tag{4.8}$$

As a note, often you will read in food labels the caloric content of certain foods. In these labels, they use the unit Calorie, with capitalized C, that is not the same as a calorie. One Calorie represents a kilocalorie, and contains 1000calories.

Sample Problem 12

Convert the following energy values:

(a) 50000 cal to Kcal (b) 48000 J to cal

SOLUTION

We will use the conversion factor for kilo and the relationship between calorie and joule:

(a) $50000\mathrm{cal} \times \frac{1\mathrm{kcal}}{1000\mathrm{cal}} = 50\mathrm{kcal}$; (b) $48000\mathrm{J} \times \frac{1\mathrm{cal}}{4.184\mathrm{J}} = 11472.2\mathrm{cal}$.

STUDY CHECK

Convert the following energy units:

(a) 200 cal to Kcal (b) 7000 J to cal

Temperature Temperature indicates how hot or cold a substance is compared to another substance. Heat always flows from a substance with a higher temperature to a substance with a lower temperature until the temperatures of both are the same. When you drink hot coffee or touch a hot pan, heat flows to your mouth or hand, which is at a lower temperature. When you touch an ice cube, it feels cold because heat flows from your hand to the colder ice cube. Three units of temperature often employed are celsius (°C, T_C), Fahrenheit (°F, T_F) or Kelvin (K, T_K). If you need to convert temperature units from Fahrenheit to celsius or from celsius to Fahrenheit you need to

use the formulas below:

$$\boxed{T_F = 1.8T_C + 32} \qquad \boxed{T_F = 1.8T_K - 459.4} \qquad \boxed{T_K = T_C + 273} \tag{4.9}$$

Sample Problem 13

Convert 25 °C to °F.

SOLUTION

1. **Step one:** list of the given variables.

	Given	Asking
Analyze the Problem	$T_c = 25°C$	T_F

2. **Step two:** use the formula $T_F = 1.8T_C + 32$ to convert from °C to °F.

$$T_F = 1.8\,T_c + 32$$

25°

3. **Step three:** solve for $T_F = 1.8 \times 25 + 32 = 77°F$.

STUDY CHECK

Convert 200°C to K.

4.7 The first law of thermodynamics. From energy to temperature

Materials can absorb heat and receive work. On one hand, think about a pizza in your over, or a cup of milk in the microwaves. These substances receive heat from the oven or in form of microwaves and they become hot. Heat transforms in an increase of temperature. On the other hand, if you hammer a wall, the wall receives work from you and this work is translated to energy as the wall may break. Work and heat are both combined in a property called internal energy, E. Heat transform in a temperature change. Some substances like metals are able to increase its temperature very quickly with a small amount of heat received, whereas others need a larger amount of heat to rise up its temperature. Think about why you use oil to deep fried food? Why not using water? First of all, oil can rise its temperature very quickly and on top of that it does not boil easily.

`Heat capacity` The heat capacity c of a material is defined as:

$$c = \frac{\text{heat adsorbed}}{\text{temperature increase}} \tag{4.10}$$

The heat capacity is a characteristic property of each material that indicates the energy required to rise its temperature and can be expressed in cal/°C or J/°C units. As this property depends on the amount of matter, often times the heat capacity is expressed

per mass as the specific heat capacity (c_e) or mole unit as the molar heat capacity c_m. For example, the specific heat of water is $1\text{cal}/\text{g}^\circ\text{C}$ that is the same as $4.184\text{J}/\text{g}^\circ\text{C}$. That means that we need to give 1 calorie in order to warm up one gram of water $1^\circ C$. Similarly, the specific heat of aluminum, a metal, is $0.2\text{cal}/\text{g}^\circ\text{C}$ or $0.89\text{J}/\text{g}^\circ\text{C}$; that means the energy needed to rise the temperature of an aluminum gram is 0.2 calories of 0.89 J. Mind the difference between these two values: we need to give 1 cal in order to increase the temperature of a gram of water in $1^\circ C$, whereas we need to give 0.2 cal in order to increase the temperature of a gram of aluminum in $1^\circ C$. Why are these two numbers so different? The answer is because water and aluminum are different materials. Normally metals warp up very easily, that is, they need less heat to increase their temperature, whereas liquids need more heat to increase their temperature. That is why pans and cooking pots tend to be metallic. Mind the specific heat if water is a well know value that you need to be familiar with:

$$\boxed{c_e^{\text{H}_2\text{O}} = 4.184\text{J}/\text{g}^\circ\text{C} \qquad or \qquad c_e^{\text{H}_2\text{O}} = 1\text{cal}/\text{g}^\circ\text{C}} \tag{4.11}$$

Table 4.2 Values of specific heat for different materials

Material	Specific heat (J/g°C)	Material	Specific heat (J/g°C)
$H_2O_{(l)}$	4.184	$Fe_{(s)}$	0.444
ethyl alcohol$_{(l)}$	2.460	$Au_{(s)}$	0.129
vegetable oil$_{(l)}$	1.790	$Cu_{(s)}$	0.385
$NH_{3(l)}$	4.700	$H_2O_{(s)}$	2.010
Dry Air$_{(g)}$	1.0035	$CO_{2(g)}$	0.839

`Heat` When a materials receives heat, that heat normally becomes temperature as the temperature of the material increases. For example, if you warm milk in a microwaves, the milk's temperature increases from room temperature (25°C) to a higher temperature. How to estimate the temperature increase given the heat received? Or how to estimate the heat needed to increase the temperature of an object? We can use the following formula:

$$\boxed{Q = m \cdot c_e \cdot (T_f - T_i)} \tag{4.12}$$

where:

Q is the amount of heat received, either in cal or J.

m is the mass of material in grams

c_e is the specific heat of the material (in $\text{cal}/\text{g}^\circ\text{C}$ or $\text{J}/\text{g}^\circ\text{C}$)

$T_f - T_i = \Delta T$, is the temperature change from the initial to the final temperature

A system can receive or give away heat and this is indicated by the sign of Q. The sign convention for heat is:

$Q > 0$ the system receives heat $\quad and \quad$ $Q < 0$ the system gives away heat

Sample Problem 14

How many calories are absorbed by a 45.2g piece of aluminum ($c_e = 0.214\frac{cal}{g°C}$) if its temperature rises from 25°C to 50°C.

SOLUTION

1. **Step one:** list of the given variables.

	Given	Asking
Analyze the Problem	$c_e = 0.214\frac{cal}{g°C}$ $m = 45.2g$ $T_{initial} = 25°C$ $T_{final} = 50°C$	Q

2. **Step two:** use the formula $Q = m \cdot c_e \cdot (T_{final} - T_{initial})$ to temperature increase to heat absorbed:

$$Q = \underbrace{m}_{45.2g} \cdot \underbrace{c_e}_{0.214\frac{cal}{g \cdot °C}} \cdot (\overbrace{T_{final}}^{25°C} - \underbrace{T_{initial}}_{50°C})$$

3. **Step three:** solve $Q = 45.2 \cdot 0.214 \cdot (50 - 25) = 241.82cal$.

STUDY CHECK

How many calories are absorbed by 100g of Gold ($c_e = 0.0308\frac{cal}{g°C}$) if its temperature rises from 25°C to 100°C.

In the previous example you needed to convert temperature into heat. In the next example, the heat is given and you need to calculate the final temperature of an object after it receives a certain amount of heat.

Sample Problem 15

A 50g piece of aluminum ($c_e = 0.214\frac{cal}{g \cdot °C}$) initially at 25°C absorbs 100cal. Calculate the final temperature of the aluminum piece.

SOLUTION

1. **Step one:** list of the given variables.

	Given	Asking
Analyze the Problem	$c_e = 0.214\frac{cal}{g \cdot °C}$ $m = 50g$ $T_{initial} = 25°C$ $Q = 100cal$	T_{final}

2. **Step two:** use the formula $Q = m \cdot c_e \cdot (T_{final} - T_{initial})$ that converts

temperature increase to heat absorbed:

$$\underbrace{\mathrm{Q}}_{100cal} = \underbrace{\mathrm{m}}_{50g} \cdot \underbrace{c_e}_{0.214\frac{cal}{g\cdot°C}} \cdot (\overbrace{T_{final} - T_{initial}}^{25°C})$$

3 **Step three:** solve $100 = 50 \cdot 0.214 \cdot (T_{final} - 25)$ for T_{final}:

$$\begin{aligned}
100 &= 50 \cdot 0.214 \cdot (T_{final} - 25) && \text{divide by 50 in both sides} \\
\frac{100}{50} &= 0.214 \cdot (T_{final} - 25) && \text{divide by 0.214 in both sides} \\
\frac{100}{50 \cdot 0.214} &= (T_{final} - 25) \\
9.34 &= (T_{final} - 25) && \text{add 25 in both sides} \\
9.34 + 25 &= T_{final} \\
34.34 &= T_{final}
\end{aligned}$$

The final temperature of the aluminum piece is $34.34°C$.

STUDY CHECK

A 200g piece of iron ($c_e = 0.1\frac{cal}{g\cdot°C}$) initially at 15°C absorbs 1000cal. Calculate the final temperature of the metal piece.